

Hongming Zhang

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Bio

Research Scientist at FAIR, Meta, working on post-training for the agentic capabilities of the Muse Spark model, with a focus on tool use, self-evolving agents, and structured reasoning models. Previously Senior Researcher and Research Lead at Tencent AI Lab, and Research Scholar at the University of Pennsylvania working with Prof. Dan Roth. Ph.D. in Computer Science from HKUST, advised by Prof. Yangqiu Song, and previously earned M.Phil. and Bachelor's degrees from HKUST.

Work Experience

META Foundational AI Research (FAIR)

RESEARCH SCIENTIST

Bellevue, WA

Jul. 2025 - Present

- Developed the tool-use post-training pipeline for the Muse Spark model, including the agentic RL infrastructure and the self-evolving data pipeline for model training.
- Contributed to Muse Spark's SOTA performance on MCP Atlas (**82.2%**) and near-SOTA performance on τ^2 -Bench Telecom (**91.5%** with original prompt, reported by Artificial Analysis).
- Founding member of the tool-use workstream, developing the model's tool-use agentic capabilities and conducting research on self-evolving agents and structured reasoning models.

Tencent AI Lab

SENIOR RESEARCHER -> RESEARCH LEAD

Bellevue, WA

Nov. 2021 - Jul. 2025

- Led a team of ten on frontier research in self-evolving LLM agents from Aug. 2023 to Jul. 2025; developed and released **Cognitive Kernel**, a **general-purpose proactive** LLM agent system for real-world interaction.
- Conducted research on **self-evolving agents**, **structured reasoning models**, and agent training for real-world interaction, with publications at ICLR, ICML, NeurIPS, ACL, and other top-tier conferences.

Education

HKUST (Hong Kong University of Science and Technology)

PH.D. IN COMPUTER SCIENCE

Hong Kong, China

Sep. 2018 - Nov. 2021

- Supervisor: Prof. Yangqiu Song
- Research topics: **commonsense reasoning**, **selectional preference**, and **coreference resolution**

University of Pennsylvania

VISITING SCHOLAR

Philadelphia, PA

Jan. 2020 - Nov. 2021

- Supervisor: Prof. Dan Roth
- Research topics: **commonsense reasoning** and **information extraction**

HKUST (Hong Kong University of Science and Technology)

M.PHIL. IN TLE (TECHNOLOGY, LEADERSHIP, AND ENTREPRENEURSHIP)

Hong Kong, China

Sep. 2016 - Aug. 2018

- Supervisor: Prof. Yangqiu Song
- Major in Computer Science; research on **graph neural networks** and **information retrieval**

HKUST (Hong Kong University of Science and Technology)

B.E. IN ELECTRONIC AND COMPUTER ENGINEERING

Hong Kong, China

Sep. 2012 - Aug. 2016

- Two minor degrees in Information Technology and Entrepreneurship
- Cumulative grade average: 3.91/4.30; ranking in major: 3/74
- Academic Achievement Medal, the highest academic award for undergraduates at HKUST

Selected Publications

- **Cognitive kernel: An open-source agent system towards generalist autopilots.** Hongming Zhang, Xiaoman Pan, Hongwei Wang, Kaixin Ma, Wenhao Yu, and Dong Yu. (NAACL 2025 Demo)
 - Create an open-source proactive agent system for generalist autopilot tasks.
- **R-Zero: Self-Evolving Reasoning LLM from Zero Data.** Chengsong Huang, Wenhao Yu, Xiaoyang Wang, Hongming Zhang, Zongxia Li, Ruosen Li, Jiaxin Huang, Haitao Mi, and Dong Yu. (ICLR 2026)
 - Develop self-evolving reasoning with zero initial data through reinforcement learning.
- **Parallel-R1: Towards Parallel Thinking via Reinforcement Learning.** Tong Zheng, Hongming Zhang, Wenhao Yu, Xiaoyang Wang, He Xing, Runpeng Dai, Rui Liu, Huiwen Bao, Chengsong Huang, Heng Huang, and Dong Yu. (ICLR 2026)
 - Develop reinforcement learning for parallel thinking in large reasoning models.

Program Committees

- **Journal Editorial Board:** Semantic Web Journal special edition on Commonsense Knowledge and Reasoning; Frontiers in Big Data.
- **Journal Reviewer:** Applied Network Science; Computer Science and Language; IEEE Transactions on Neural Networks and Learning Systems; NeuralComputing.
- **Area Chair and Conference Committee:** ICLR'24-26; ICML'25; NeurIPS'23-26; COLM'24-25; AACL'23; ACL'21-24; EMNLP'21-24; CVPR 22-24; NAACL'21-24; IJCAI'21-23; AAAI'21-23; COLING'20; AACL'20; CIKM'19; AKBC'21.
- **Others:** IJCAI 2019 WebMaster.

Honors & Awards

2024	Outstanding Paper , Transactions of Machine Learning Research	Bellevue, WA, U.S.A
2023	Outstanding Paper , EMNLP	Singapore
2022	Engineering PhD Research Excellence Finalist Award , HKUST	Hong Kong, China
2020	SENG Academic Award for Continuing PhD students , HKUST	Hong Kong, China
2019	Microsoft Research Asia Fellowship , Microsoft Research Asia	Beijing, China
2019	Tencent Rhino-Bird Scholarship , Tencent	Shenzhen, China
2018	Hong Kong PhD Fellowship , Hong Kong	Hong Kong, China
2018	Excellent Research Award , HKUST	Hong Kong, China
2016	Academic Achievement Medal , HKUST	Hong Kong, China
2014	Academic Excellence Award , HKUST	Hong Kong, China

All Peer-reviewed Publications

2026

- Fengyu Cai, Tong Chen, Xinran Zhao, Sihao Chen, Hongming Zhang, Tongshuang Wu, Iryna Gurevych, and Heinz Koepl. Revela: Dense Retriever Learning via Language Modeling. (ICLR 2026 Oral)
- Chengsong Huang, Wenhao Yu, Xiaoyang Wang, Hongming Zhang, Zongxia Li, Ruosen Li, Jiaxin Huang, Haitao Mi, and Dong Yu. R-Zero: Self-Evolving Reasoning LLM from Zero Data. (ICLR 2026)
- Tong Zheng, Hongming Zhang, Wenhao Yu, Xiaoyang Wang, He Xing, Runpeng Dai, Rui Liu, Huiwen Bao, Chengsong Huang, Heng Huang, and Dong Yu. Parallel-R1: Towards Parallel Thinking via Reinforcement Learning. (ICLR 2026)
- Ce Zhang, Kaixin Ma, Tianqing Fang, Wenhao Yu, Hongming Zhang, Zhisong Zhang, Haitao Mi, and Dong Yu. VScan: Rethinking Visual Token Reduction for Efficient Large Vision-Language Models. (TMLR 2026)
- Zhisong Zhang, Tianqing Fang, Kaixin Ma, Wenhao Yu, Hongming Zhang, Haitao Mi, and Dong Yu. WebRollback: Enhancing Web Agents with Explicit Rollback Mechanisms. (EACL 2026 Short)

2025

- Hongming Zhang, Xiaoman Pan, Hongwei Wang, Kaixin Ma, Wenhao Yu, and Dong Yu. Cognitive Kernel: An Open-source Agent System towards Generalist Autopilots. (NAACL 2025 Demo)
- Liqiang Jing, Zhehui Huang, Xiaoyang Wang, Wenlin Yao, Wenhao Yu, Kaixin Ma, Hongming Zhang, Xinya Du, and Dong Yu. DSbench: How Far Are Data Science Agents from Becoming Data Science Experts? (ICLR 2025)
- Siru Ouyang, Wenhao Yu, Kaixin Ma, Zilin Xiao, Zhihan Zhang, Mengzhao Jia, Jiawei Han, Hongming Zhang, and Dong Yu. RepoGraph: Enhancing AI Software Engineering with Repository-level Code Graph. (ICLR 2025)

- Mengzhao Jia, Wenhao Yu, Kaixin Ma, Tianqing Fang, Zhihan Zhang, Siru Ouyang, **Hongming Zhang**, Dong Yu, and Meng Jiang. Leopard: A Vision Language Model for Text-Rich Multi-Image Tasks. (TMLR 2025)
- Tianqing Fang, **Hongming Zhang**, Zhisong Zhang, Kaixin Ma, Wenhao Yu, Haitao Mi, and Dong Yu. WebEvolver: Enhancing Web Agent Self-Improvement with Co-evolving World Model. (EMNLP 2025)
- Jun-Yu Ma, Tianqing Fang, Zhisong Zhang, **Hongming Zhang**, Haitao Mi, and Dong Yu. Recall with Reasoning: Chain-of-Thought Distillation for Mamba’s Long-Context Memory and Extrapolation. (EMNLP 2025)
- Ran Xu, Kaixin Ma, Wenhao Yu, **Hongming Zhang**, Joyce C. Ho, Carl Yang, and Dong Yu. Retrieval-augmented GUI Agents with Generative Guidelines. (EMNLP 2025)
- Minda Hu, Tianqing Fang, Jianshu Zhang, Jun-Yu Ma, Zhisong Zhang, Jingyan Zhou, **Hongming Zhang**, Haitao Mi, Dong Yu, and Irwin King. WebCoT: Enhancing Web Agent Reasoning by Reconstructing Chain-of-Thought in Reflection, Branching, and Rollback. (Findings of EMNLP 2025)
- Zhaowei Wang, **Hongming Zhang**, Tianqing Fang, Ye Tian, Yue Yang, Kaixin Ma, Xiaoman Pan, Yangqiu Song, and Dong Yu. DivScene: Towards Open-Vocabulary Object Navigation with Large Vision Language Models in Diverse Scenes. (Findings of EMNLP 2025)

2024

- With many other collaborators. Beyond the Imitation Game: Quantifying and Extrapolating the Capabilities of Language Models. (TMLR 2024, Outstanding Paper)
- Wenhao Yu, **Hongming Zhang**, Xiaoman Pan, Peixin Cao, Kaixin Ma, Jian Li, Hongwei Wang, and Dong Yu. Chain-of-Note: Enhancing Robustness in Retrieval-Augmented Language Models. (EMNLP 2024)
- Fengyu Cai, Xinran Zhao, Tong Chen, Sihao Chen, **Hongming Zhang**, Iryna Gurevych, and Heinz Koepl. MixGR: Enhancing Retriever Generalization for Scientific Domain through Complementary Granularity. (EMNLP 2024)
- Tong Chen, Hongwei Wang, Sihao Chen, Wenhao Yu, Kaixin Ma, Xinran Zhao, **Hongming Zhang**, and Dong Yu. Dense X Retrieval: What Retrieval Granularity Should We Use? (EMNLP 2024)
- Ruixin Hong, **Hongming Zhang**, Xiaoman Pan, Dong Yu, and Changshui Zhang. Abstraction-of-Thought Makes Language Models Better Reasoners. (Findings of EMNLP 2024)
- Chunkit Chan, Jiayang Cheng, Yauwai Yim, Zheyang Deng, Wei Fan, Haoran Li, Xin Liu, **Hongming Zhang**, Weiqi Wang, and Yangqiu Song. NegotiationToM: A Benchmark for Stress-testing Machine Theory of Mind on Negotiation Surrounding. (Findings of EMNLP 2024)
- Zhaowei Wang, Wei Fan, Qing Zong, **Hongming Zhang**, Sehyun Choi, Tianqing Fang, Xin Liu, Yangqiu Song, Ginny Wong, and Simon See. AbsInstruct: Eliciting Abstraction Ability from LLMs through Explanation Tuning with Plausibility Estimation. (ACL 2024)
- Yinya Huang, Ruixin Hong, **Hongming Zhang**, Wei Shao, Zhicheng Yang, Dong Yu, Changshui Zhang, Xiaodan Liang, and Linqi Song. CLOMO: Counterfactual Logical Modification with Large Language Models. (ACL 2024)
- Hongliang He, Wenlin Yao, Kaixin Ma, Wenhao Yu, Yong Dai, **Hongming Zhang**, Zhenzhong Lan, and Dong Yu. WebVoyager: Building an End-to-End Web Agent with Large Multimodal Models. (ACL 2024)
- Xinran Zhao, **Hongming Zhang**, Xiaoman Pan, Wenlin Yao, Dong Yu, Tongshuang Wu, and Jianshu Chen. Fact-and-Reflection (FaR) Improves Confidence Calibration of Large Language Models. (Findings of ACL 2024)
- Fengyu Cai, Xinran Zhao, **Hongming Zhang**, Iryna Gurevych, and Heinz Koepl. GEOHard: Towards Measuring Class-wise Hardness through Modelling Class Semantics. (Findings of ACL 2024)
- Ruixin Hong, **Hongming Zhang**, Xinyu Pang, Dong Yu, and Changshui Zhang. A Closer Look at the Self-Verification Abilities of Large Language Models in Logical Reasoning. (NAACL 2024)
- Sihao Chen, **Hongming Zhang**, Tong Chen, Ben Zhou, Wenhao Yu, Dian Yu, Baolin Peng, Hongwei Wang, Dan Roth, and Dong Yu. Sub-Sentence Encoder: Contrastive Learning of Propositional Semantic Representations. (NAACL 2024)
- Hangfeng He, **Hongming Zhang**, and Dan Roth. SocREval: Large Language Models with the Socratic Method for Reference-free Reasoning Evaluation. (Findings of NAACL 2024)
- Tianqing Fang, Zhaowei Wang, Wenxuan Zhou, **Hongming Zhang**, Yangqiu Song, and Muhao Chen. Getting Sick After Seeing a Doctor? Diagnosing and Mitigating Knowledge Conflicts in Event Temporal Reasoning. (Findings of NAACL 2024)
- Zhaowei Wang, Haochen Shi, Weiqi Wang, Tianqing Fang, **Hongming Zhang**, Sehyun Choi, Xin Liu, and Yangqiu Song. AbsPyramid: Benchmarking the Abstraction Ability of Language Models with a Unified Entailment Graph. (Findings of NAACL 2024)
- Nan Xu, **Hongming Zhang**, and Jianshu Chen. CEO: Corpus-based Open-Domain Event Ontology Induction. (Findings of EACL 2024)
- Tianqing Fang, Wenxuan Zhou, Fangyu Liu, **Hongming Zhang**, Yangqiu Song, and Muhao Chen. On-the-fly Denoising for Data Augmentation in Natural Language Understanding. (Findings of EACL 2024)

- Haoyu Wang, **Hongming Zhang**, Kaiqiang Song, Dong Yu, and Dan Roth. Event Semantic Classification in Context. (Findings of EACL 2024)

2023

- Haoyu Wang, **Hongming Zhang**, Yueguan Wang, Yuqian Deng, Muhao Chen, and Dan Roth. Are All Steps Equally Important? Benchmarking Essentiality Detection of Events. (EMNLP 2023)
- Hongwei Wang, **Hongming Zhang**, and Dong Yu. On the Dimensionality of Sentence Embeddings. (Findings of EMNLP 2023)
- Keming Lu, Xiaoman Pan, Kaiqiang Song, **Hongming Zhang**, Dong Yu, and Jianshu Chen. PIVOINE: Instruction Tuning for Open-world Information Extraction. (Findings of EMNLP 2023)
- Jiayang Cheng, Lin Qiu, Tsz Ho Chan, Tianqing Fang, Weiqi Wang, Chunkit Chan, Qipeng Guo, **Hongming Zhang**, Yangqiu Song, Yue Zhang, and Zheng Zhang. STORYANALOGY: Deriving Story-level Analogies from Large Language Models to Unlock Analogical Understanding. (EMNLP 2023)
- James Y. Huang, Wenlin Yao, Kaiqiang Song, **Hongming Zhang**, Muhao Chen, and Dong Yu. Bridging Continuous and Discrete Spaces: Interpretable Sentence Representation Learning via Compositional Operations. (EMNLP 2023 Outstanding Paper)
- Xinran Zhao, **Hongming Zhang**, Xiaoman Pan, Wenlin Yao, Dong Yu, and Jianshu Chen. Thrust: Adaptively Propels Large Language Models with External Knowledge. (NeurIPS 2023)
- Xiang Li, Jiangwei Yu, Xinran Zhao, **Hongming Zhang**, and Yu-Xiong Wang. Video State-changing Object Segmentation. (ICCV 2023)
- Ruixin Hong, **Hongming Zhang**, Hong Zhao, Dong Yu, and Changshui Zhang. Faithful Question Answering with Monte-Carlo Planning. (ACL 2023)
- Zhaowei Wang, Quyet V. Do, **Hongming Zhang**, Jiayao Zhang, Weiqi Wang, Tianqing Fang, Yangqiu Song, Ginny Y. Wong, and Simon See. COLA: Contextualized Commonsense Causality Reasoning from the Causal Inference Perspective. (ACL 2023)
- Xiaoman Pan, Wenlin Yao, **Hongming Zhang**, Dian Yu, Dong Yu, and Jianshu Chen. Knowledge-in-Context: Towards Knowledgeable Semi-Parametric Language Models. (ICLR 2023)
- Haoyu Wang, **Hongming Zhang**, Yuqian Deng, Jacob R. Gardner, Muhao Chen, and Dan Roth. Extracting or Guessing? Improving Faithfulness of Event Temporal Relation Extraction. (EACL 2023)
- Wenlin Yao, Lifeng Jin, **Hongming Zhang**, Xiaoman Pan, Kaiqiang Song, Dian Yu, Dong Yu, and Jianshu Chen. How do Words Contribute to Sentence Semantics? Revisiting Sentence Embeddings with a Perturbation Method. (Findings of EACL 2023)
- **Hongming Zhang**, Yintong Huo, Yanai Elazar, Yangqiu Song, Yoav Goldberg, and Dan Roth. CIKQA: Learning Commonsense Inference with a Unified Knowledge-in-the-loop QA Paradigm. (Findings of EACL 2023)
- Zizheng Lin, **Hongming Zhang**, and Yangqiu Song. Global Constraints with Prompting for Zero-Shot Event Argument Classification. (Findings of EACL 2023)

2022

- **Hongming Zhang**, Xin Liu, Haojie Pan, Haowen Ke, Tianqing Fang, Jiefu Ou, and Yangqiu Song. ASER: Towards Large-scale Commonsense Knowledge Acquisition via Higher-order Selectional Preference over Eventualities. (Artificial Intelligence 2022)
- Xintong Yu, **Hongming Zhang**, Ruixin Hong, Yangqiu Song, and Changshui Zhang. CD-PCR: Improving Visual Dialog with Pronoun Coreference Resolution. (Pattern Recognition 2022)
- Ruixin Hong, **Hongming Zhang**, Xintong Yu, and Changshui Zhang. Learning Event Extraction From a Few Guideline Examples. (IEEE/ACM Transactions on Audio, Speech, and Language Processing 2022)
- Yinya Huang, **Hongming Zhang**, Ruixin Hong, Xiaodan Liang, Changshui Zhang, and Dong Yu. MetaLogic: Logical Reasoning Explanations with Fine-Grained Structure. (EMNLP 2022)
- Zhaowei Wang, **Hongming Zhang**, Tianqing Fang, Yangqiu Song, Ginny Y. Wong, and Simon See. SubeventWriter: Iterative Sub-event Sequence Generation with Coherence Controller. (EMNLP 2022)
- Fei Wang, Kaiqiang Song, **Hongming Zhang**, Lifeng Jin, Sangwoo Cho, Wenlin Yao, Xiaoyang Wang, Muhao Chen, and Dong Yu. Saliency Allocation as Guidance for Abstractive Summarization. (EMNLP 2022)
- Yue Yang, Wenlin Yao, **Hongming Zhang**, Xiaoyang Wang, Dong Yu, and Jianshu Chen. Z-LaVI: Zero-Shot Language Solver Fueled by Visual Imagination. (EMNLP 2022)
- **Hongming Zhang**, Wenlin Yao, and Dong Yu. Efficient Zero-shot Event Extraction with Context-Definition Alignment. (Findings of EMNLP 2022)

- Tianqing Fang, Quyet V. Do, **Hongming Zhang**, Yangqiu Song, Ginny Y. Wong, and Simon See. PseudoReasoner: Leveraging Pseudo Labels for Commonsense Knowledge Base Population. (Findings of EMNLP 2022)
- Ying Su, Zihao Wang, Tianqing Fang, **Hongming Zhang**, Yangqiu Song, and Tong Zhang. MICO: A Multi-alternative Contrastive Learning Framework for Commonsense Knowledge Representation. (Findings of EMNLP 2022)
- Ying Su, **Hongming Zhang**, Yangqiu Song, and Tong Zhang. Multilingual Word Sense Disambiguation with Unified Sense Representation. (COLING 2022)
- Jiayao Zhang, **Hongming Zhang**, Weijie Su, and Dan Roth. Causal Inference Principles for Reasoning about Commonsense Causality. (ICML 2022)
- Hantian Ding, Jinrui Yang, Yuqian Deng, **Hongming Zhang**, and Dan Roth. Towards Open-Domain Topic Classification. (NAACL 2022 Demo)
- Ruixin Hong, **Hongming Zhang**, Xintong Yu, and Changshui Zhang. METGEN: A Module-based Entailment Tree Generation Framework for Answer Explanation. (Findings of NAACL 2022)
- Jiaxin Bai, Zihao Wang, **Hongming Zhang**, and Yangqiu Song. Query2Particles: Knowledge Graph Reasoning with Particle Embeddings. (Findings of NAACL 2022)
- Xinran Zhao, **Hongming Zhang**, and Yangqiu Song. PCR4ALL: A Comprehensive Evaluation Benchmark for Pronoun Coreference Resolution in English. (LREC 2022)
- Changlong Yu, **Hongming Zhang**, Yangqiu Song, and Wilfred Ng. CoCoLM: Complex Commonsense Enhanced Language Model. (Findings of ACL 2022)
- Ying Su, **Hongming Zhang**, Yangqiu Song, and Tong Zhang. Rare and Zero-shot Word Sense Disambiguation using Z-Reweighting. (ACL 2022)
- Yintong Huo, Yuxin Su, **Hongming Zhang**, and Michael Lyu. ARCLIN: Automated API Mention Resolution for Unformatted Texts. (ICSE 2022)

2021

- Haoyu Wang, **Hongming Zhang**, Muhao Chen, and Dan Roth. Learning Constraints and Descriptive Segmentation for Subevent Detection. (EMNLP 2021)
- Yanai Elazar, **Hongming Zhang**, Yoav Goldberg, and Dan Roth. Back to Square One: Bias Detection, Training and Commonsense Disentanglement in the Winograd Schema. (EMNLP 2021)
- Xintong Yu, **Hongming Zhang**, Yangqiu Song, Changshui Zhang, Kun Xu, and Dong Yu. Exophoric Pronoun Resolution in Dialogues with Topic Regularization. (EMNLP 2021)
- Tianqing Fang, Weiqi Wang, Sehyun Choi, Shibo Hao, **Hongming Zhang**, Yangqiu Song, and Bin He. Benchmarking Commonsense Knowledge Base Population with an Effective Evaluation Dataset. (EMNLP 2021)
- Tianqing Fang, Haojie Pan, **Hongming Zhang**, Yangqiu Song, Kun Xu, and Dong Yu. Do Boat and Ocean Suggest Beach? Dialogue Summarization with External Knowledge. (AKBC 2021)
- **Hongming Zhang**, Haoyu Wang, and Dan Roth. Zero-shot Label-Aware Event Trigger and Argument Classification. (Findings of ACL 2021)
- Qing Lyu, **Hongming Zhang**, Elior Sulem, and Dan Roth. Zero-shot Event Extraction via Transfer Learning: Challenges and Insights. (ACL 2021)
- Xinran Zhao, Esin Durmus, **Hongming Zhang**, and Claire Cardie. Leveraging Topic Relatedness for Argument Persuasion. (Findings of ACL 2021)
- **Hongming Zhang**, Yintong Huo, Xinran Zhao, Yangqiu Song, and Dan Roth. Learning Contextual Causality from Time-consecutive Images. (CVPR 2021)
- Tianqing Fang, **Hongming Zhang**, Weiqi Wang, Yangqiu Song, and Bin He. DISCOS: Bridging the Gap between Discourse Knowledge and Commonsense Knowledge. (WWW 2021)
- Jiaxin Bai, **Hongming Zhang**, Yangqiu Song, and Kun Xu. Joint Coreference Resolution and Character Linking for Multi-party Conversation. (EACL 2021)

2020

- Muhao Chen, **Hongming Zhang**, Haoyu Wang, and Dan Roth. What Are You Trying to Do? Semantic Typing of Event Processes. (CoNLL 2020)
- **Hongming Zhang**, Muhao Chen, Haoyu Wang, Yangqiu Song, and Dan Roth. Analogous Process Structure Induction for Sub-event Sequence Prediction. (EMNLP 2020)
- Haoyu Wang, Muhao Chen, **Hongming Zhang**, and Dan Roth. Joint Constrained Learning for Event-Event Relation Extraction. (EMNLP 2020)

- Changlong Yu, Jialong Han, Peifeng Wang, Yangqiu Song, **Hongming Zhang**, Wilfred Ng, and Shuming Shi. When Hearst Is not Enough: Improving Hypernymy Detection from Corpus with Distributional Models. (EMNLP 2020)
- Changlong Yu, **Hongming Zhang**, Yangqiu Song, Wilfred Ng, and Lifeng Shang. Enriching Large-Scale Eventuality Knowledge Graph with Entailment Relations. (AKBC 2020)
- **Hongming Zhang**, Daniel Khashabi, Yangqiu Song, and Dan Roth. TransOMCS: From Linguistic Graphs to Commonsense Knowledge. (IJCAI 2020)
- **Hongming Zhang**, Xinran Zhao, and Yangqiu Song. WinoWhy: A Deep Diagnosis of Essential Commonsense Knowledge for Answering Winograd Schema Challenge. (ACL 2020)
- **Hongming Zhang**, Xin Liu, Haojie Pan, Yangqiu Song, and Cane Wing-Ki Leung. ASER: A Large-scale Eventuality Knowledge Graph. (WWW 2020)

2019

- **Hongming Zhang**, Jiaxin Bai, Yan Song, Kun Xu, Changlong Yu, Yangqiu Song, Wilfred Ng, and Dong Yu. Multiplex Word Embeddings for Selectional Preference Acquisition. (EMNLP 2019)
- Xintong Yu, **Hongming Zhang**, Yangqiu Song, Yan Song, and Changshui Zhang. What You See is What You Get: Visual Pronoun Coreference Resolution in Conversations. (EMNLP 2019)
- Nedjma Ousidhoum, Zizheng Lin, **Hongming Zhang**, Yangqiu Song, and Dit-Yan Yeung. Multilingual and Multi-aspect Hate Speech Analysis. (EMNLP 2019)
- **Hongming Zhang**, Hantian Ding, and Yangqiu Song. SP-10K: A Large-Scale Evaluation Set for Selectional Preference Acquisition. (ACL 2019)
- **Hongming Zhang**, Yan Song, Yangqiu Song, and Dong Yu. Knowledge-aware Pronoun Coreference Resolution. (ACL 2019)
- **Hongming Zhang**, Yan Song, and Yangqiu Song. Incorporating Context and External Knowledge for Pronoun Coreference Resolution. (NAACL 2019)

2018

- **Hongming Zhang**, Liwei Qiu, Lingling Yi, and Yangqiu Song. Scalable Multiplex Network Embedding. (IJCAI 2018)